

Pracownia Systemów i Urządzeń Sieciowych

**09.03.2023 – Podsumowanie Windows Serwer + Konfiguracja sieci
LAN**



Imię i Nazwisko

klasa/grupa/ rok szk.

Ksawery Zelek

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2.

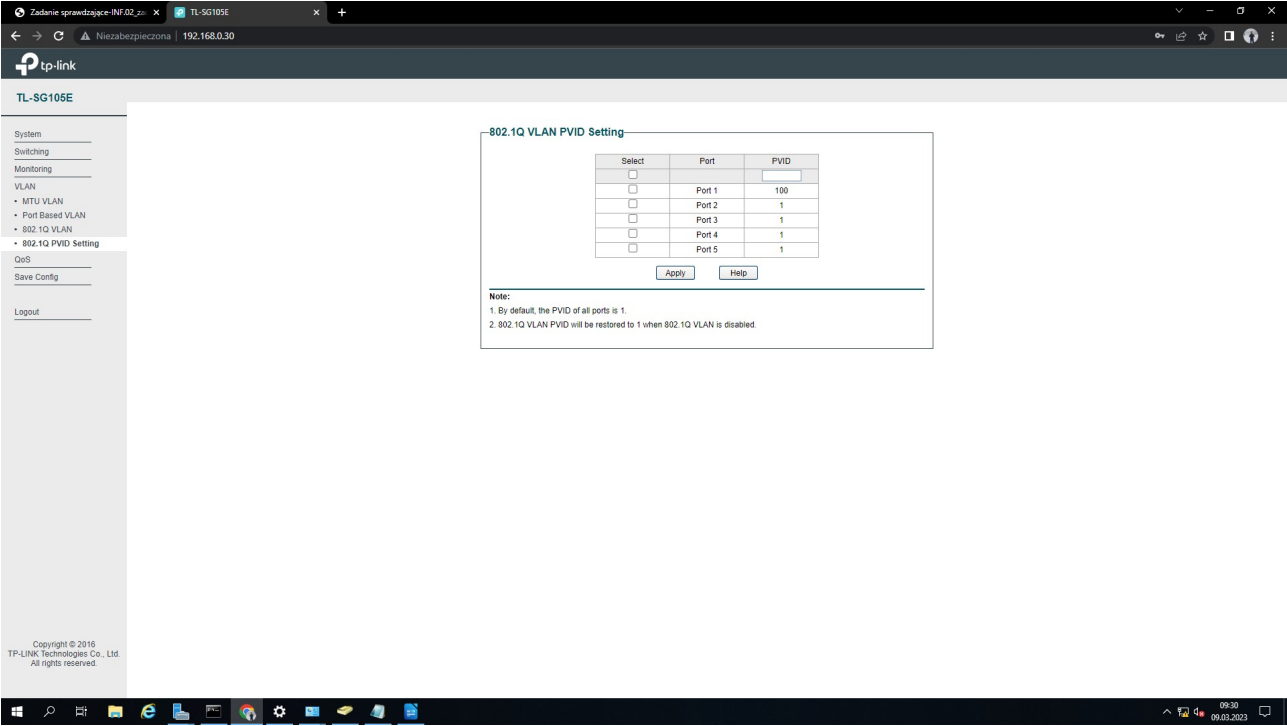
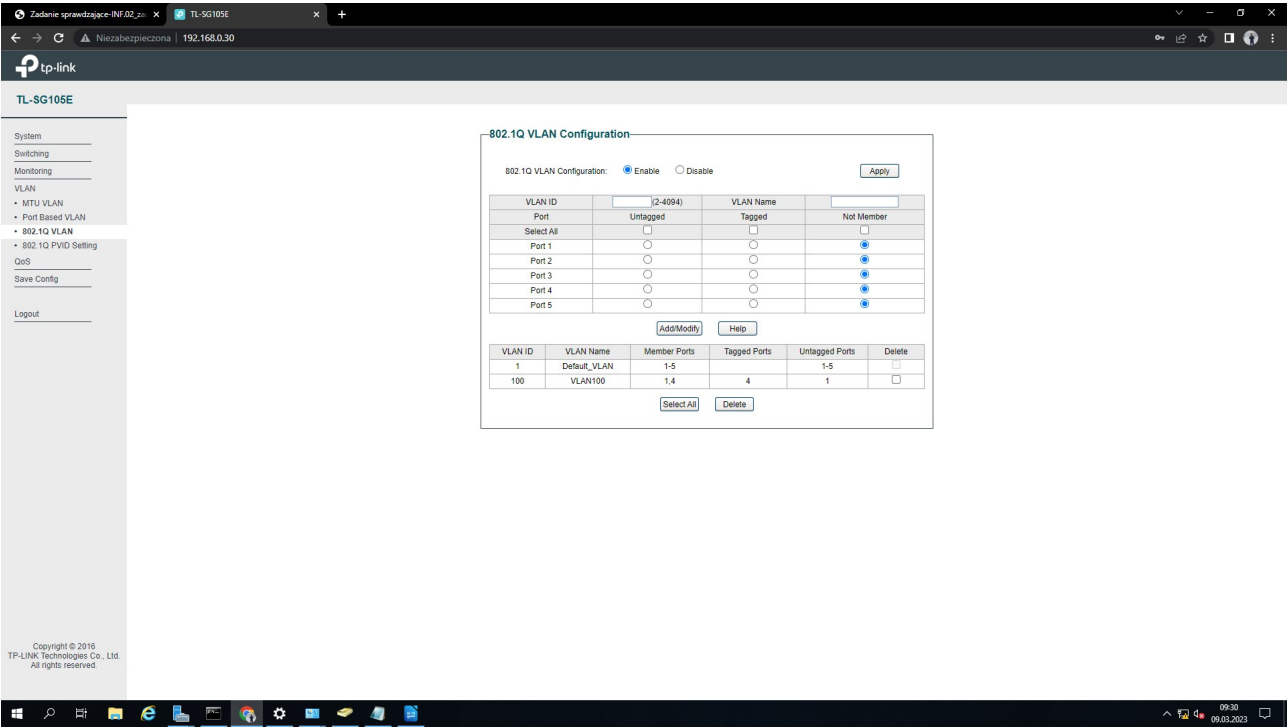
The screenshot shows the TP-Link Archer C1200 web interface. The left sidebar contains a menu with options: Stan, Sieć (selected), Internet, LAN, IPTV/VLAN, Serwer DHCP, Dynamiczny DNS, Routing zaawansowany, Tryb pracy, Sieć bezprzewodowa, Sieć dla gości, Przekierowanie portów, Ustawienia USB, Kontrola rodzicielska, QoS, and Zabezpieczenia. The main content area is titled 'LAN' and displays the following configuration: Adres MAC: B0-4E-26-D6-78-50, Adres IP: 192.168.0.103, and Maska podsieci: 255.255.255.0. A 'Zapisz' button is visible at the bottom right of the configuration area. The footer indicates the firmware version is 2.0.3 Build 20211219 Rel. 67401 (EU) and the hardware version is Archer C1200 v2.0.

The screenshot shows the TP-Link Archer C1200 web interface, specifically the 'Ustawienia' (Settings) page for DHCP. The left sidebar is identical to the previous screenshot. The main content area is titled 'Ustawienia' and displays the following configuration: Serwer DHCP: ☐ Włącz serwer DHCP, Puła adresów IP: 192.168.0.100 - 192.168.0.199, Czas przydzielenia adresu: 120 minut, Brama domyślna: 192.168.0.103 (Opcjonalnie), Preferowany DNS: (Opcjonalnie), and Alternatywny DNS: (Opcjonalnie). A 'Zapisz' button is visible at the bottom right of the configuration area. Below the settings, there is a section for 'Rezerwacja adresów' (Reservation) with a table showing reserved IP addresses. The table has columns: ID, Adres MAC, Zarezerwowany adres IP, Opis, Stan, and Zmień. Below this, there is a section for 'Lista klientów DHCP' (DHCP Clients) with a table showing the list of DHCP clients. The table has columns: ID, Nazwa Klienta, Adres MAC, Przydzielone IP, and Czas przydzielenia. The footer indicates the firmware version is 2.0.3 Build 20211219 Rel. 67401 (EU) and the hardware version is Archer C1200 v2.0.

ID	Adres MAC	Zarezerwowany adres IP	Opis	Stan	Zmień
...	...	...	...	...	...

ID	Nazwa Klienta	Adres MAC	Przydzielone IP	Czas przydzielenia
1	KsZelek-Stacja	4C-CC-6A-D7-DA-60	192.168.0.176	1:21:14
2	KsZelek-Serwer	4C-CC-6A-F5-A1-5D	192.168.0.193	0:56:8

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The screenshot shows the TP-Link T1600G-28TS web interface. The left sidebar contains a menu with options: System, Switching, VLAN, 802.1Q VLAN, MAC VLAN, Protocol VLAN, Spanning Tree, Multicast, Routing, QoS, ACL, Network Security, SNMP, LLDP, Maintenance, Save Config, Index, and Logout. The main content area is titled 'VLAN Config' and 'Port Config'. Under 'VLAN Config', there is a 'VLAN Info' section with fields for 'VLAN ID' (set to 100) and 'Name' (set to VLAN100). Below this are 'Untagged port' and 'Tagged port' sections, each with a 'UNIT' dropdown set to 'LAGS' and a grid of port selection buttons. At the bottom, there are checkboxes for 'Unselected Port(s)', 'Selected Port(s)', and 'Not Available for Selection'.

The screenshot shows the TP-Link T1600G-28TS web interface, specifically the 'Port Config' tab. The 'VLAN Port Config' section displays a table with columns: Select, Port, PVID, LAG, and VLAN. The table lists ports 1/0/1 through 1/0/15, all with PVID set to 1 and LAG set to ---. Each row has a 'Detail' link. Below the table are 'All', 'Apply', and 'Help' buttons.

Select	Port	PVID	LAG	VLAN
☐	1/0/1	1	---	Detail
☐	1/0/2	100	---	Detail
☐	1/0/3	100	---	Detail
☐	1/0/4	1	---	Detail
☐	1/0/5	1	---	Detail
☐	1/0/6	1	---	Detail
☐	1/0/7	1	---	Detail
☐	1/0/8	1	---	Detail
☐	1/0/9	1	---	Detail
☐	1/0/10	1	---	Detail
☐	1/0/11	1	---	Detail
☐	1/0/12	1	---	Detail
☐	1/0/13	1	---	Detail
☐	1/0/14	1	---	Detail
☐	1/0/15	1	---	Detail

5. Działanie VLANów

```
Administrator: Wiersz polecenia

C:\Users\Administrator>ping 192.168.0.20

Pinging 192.168.0.20 with 32 bytes of data:
Reply from 192.168.0.130: Destination host unreachable.
Reply from 192.168.0.130: Destination host unreachable.
Reply from 192.168.0.130: Destination host unreachable.
Reply from 192.168.0.130: Destination host unreachable.

Ping statistics for 192.168.0.20:
    Packets: Sent = 4, Received = 0 (0% loss),
C:\Users\Administrator>ping 192.168.0.30

Pinging 192.168.0.30 with 32 bytes of data:
Reply from 192.168.0.30: bytes=32 time=3ms TTL=64
Reply from 192.168.0.30: bytes=32 time=1ms TTL=64
Reply from 192.168.0.30: bytes=32 time=1ms TTL=64
Reply from 192.168.0.30: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.0.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms
C:\Users\Administrator>ping 192.168.0.103

Pinging 192.168.0.103 with 32 bytes of data:
Reply from 192.168.0.103: bytes=32 time<1ms TTL=64
Reply from 192.168.0.103: bytes=32 time<1ms TTL=64
Reply from 192.168.0.103: bytes=32 time<1ms TTL=64
Reply from 192.168.0.103: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.0.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Users\Administrator>ping 192.168.0.200

Pinging 192.168.0.200 with 32 bytes of data:
Reply from 192.168.0.200: bytes=32 time<1ms TTL=128
Reply from 192.168.0.200: bytes=32 time<1ms TTL=128
Reply from 192.168.0.200: bytes=32 time<1ms TTL=128
Reply from 192.168.0.200: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Users\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter WAN:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Ethernet adapter NIC2:

Administrator: Wiersz polecenia
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Administrator: Wiersz polecenia

C:\Users\Administrator>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : Ksiazek-Serwer
    Primary Dns Suffix . . . . . : Zelek.Local
    Node Type . . . . . : Hybrid
    IP Routing Enabled . . . . . : No
    WINS Proxy Enabled . . . . . : No
    DNS Suffix Search List. . . . . : Zelek.Local

Ethernet adapter WAN:

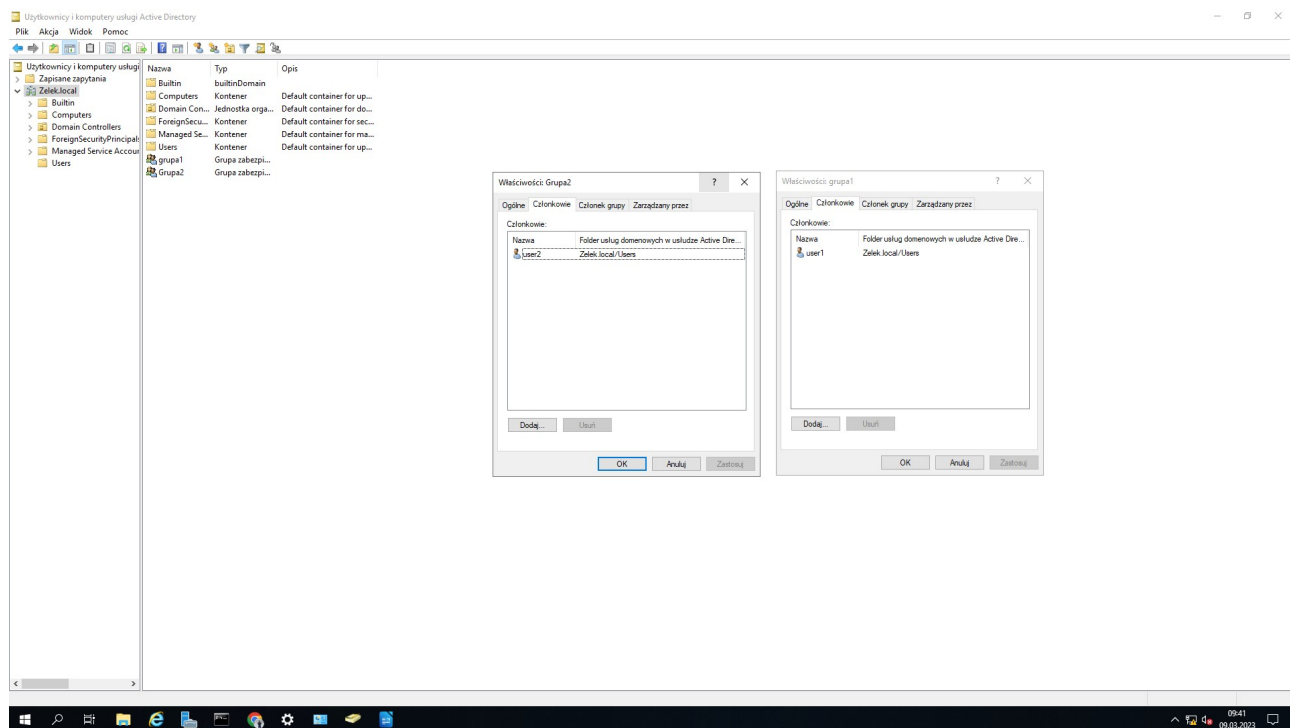
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Intel(R) Ethernet Connection I219-LM
    Physical Address. . . . . : 4C-CE-6A-F5-A1-50
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Ethernet adapter NIC2:

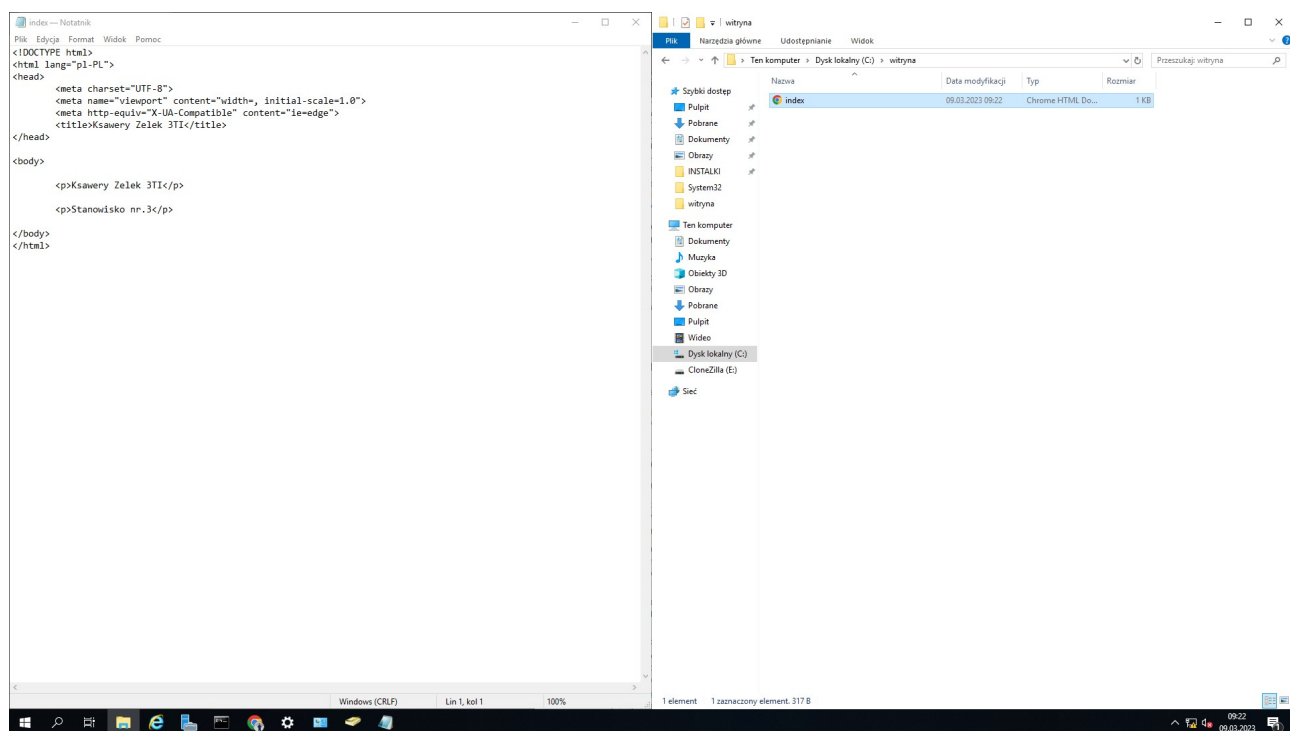
    Connection-specific DNS Suffix . :
    Description . . . . . : TP-Link Gigabit PCI Express Adapter #2
    Physical Address. . . . . : C4-6E-1F-04-9A-62
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::a0e8:84fc:c508:6e13%18(Prefered)
    IPv4 Address. . . . . : 192.168.0.130(Prefered)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.0.10
    DHCPv6 IAID . . . . . : 465858099
    DHCPv6 Client DUID. . . . . : 00-01-00-01-28-CA-60-47-00-E0-4C-68-01-66
    DNS Servers . . . . . : 192.168.0.10
    NetBIOS over Tcpip. . . . . : Enabled

C:\Users\Administrator>
```

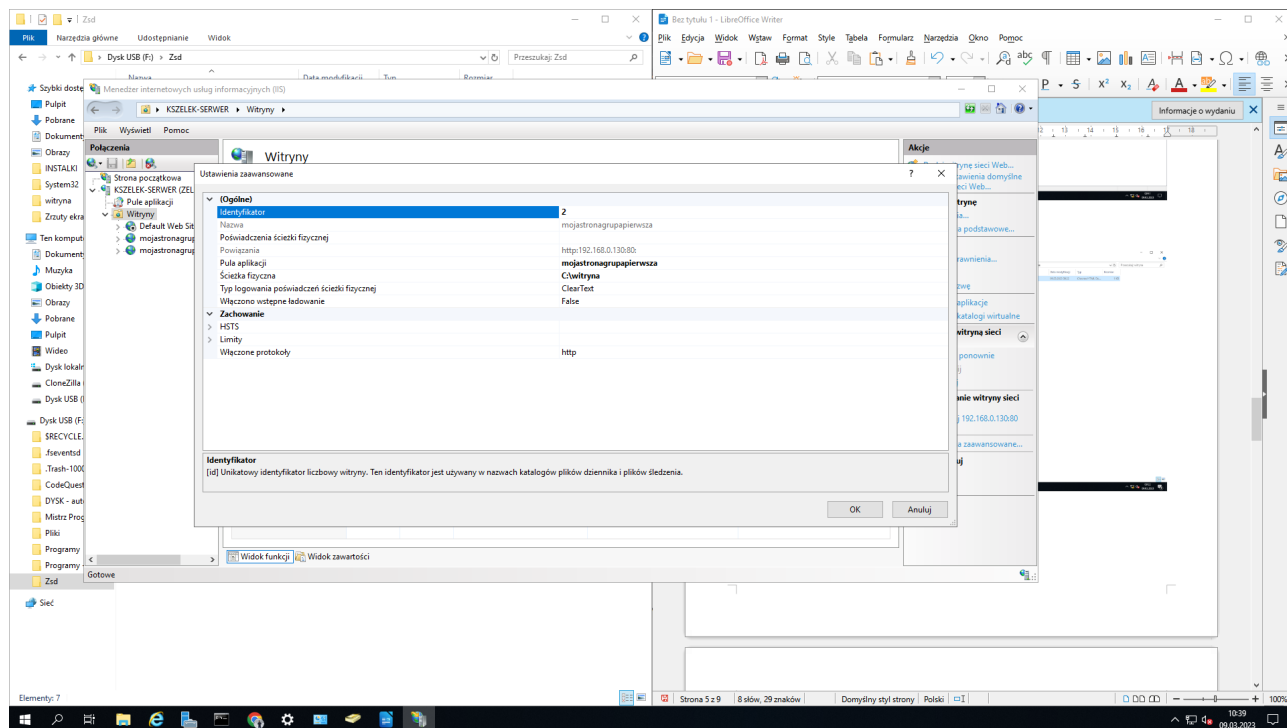
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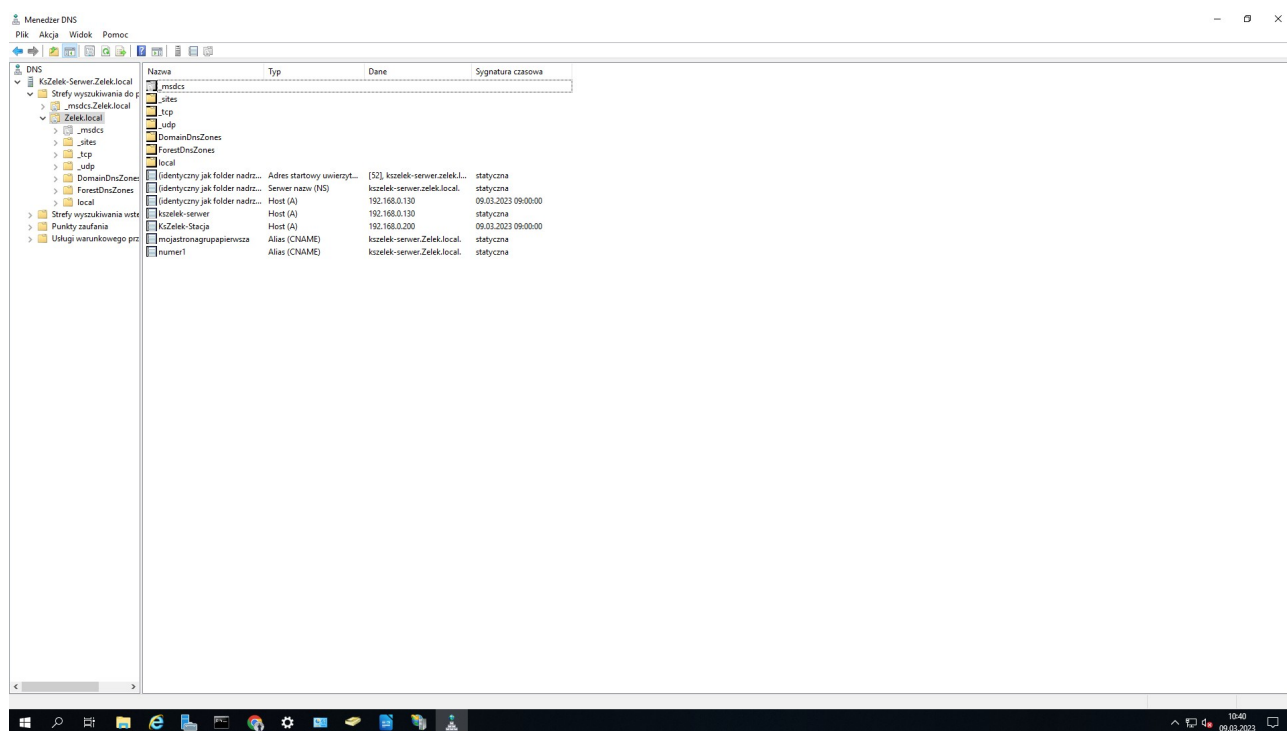
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```
Wiersz polecenia
Microsoft Windows [Version 10.0.19044.1766]
(c) Microsoft Corporation. Wszelkie prawa zastrzeżone.

C:\Users\User1>ipconfig /all

Windows IP Configuration

Host Name . . . . . : Ksiazek-Stacja
Primary Dns Suffix . . . . . : Zelek.local
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : Zelek.local

Ethernet adapter LAN2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : TP-Link Gigabit PCI Express Adapter
Physical Address. . . . . : 18-06-C7-04-48-B5
DKDP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter NIC1:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Ethernet Connection (2) I219-V
Physical Address. . . . . : 4C-CC-6A-D7-DA-60
DKDP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::887b:67d7:a951:290c%15(Preferred)
IPv4 Address. . . . . : 192.168.0.200(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.0.103
DKCPv6 IAD . . . . . : 38587498
DKCPv6 Client DUID. . . . . : 00-01-00-01-28-9B-8D-4C-CC-6A-D7-DA-60
DNS Servers . . . . . : 192.168.0.130
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\User1>
```

12 a.

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Wiersz polecenia
Microsoft Windows [Version 10.0.19044.1766]
(c) Microsoft Corporation. Wszelkie prawa zastrzeżone.

C:\Users\User1>ping 192.168.0.20

Pinging 192.168.0.20 with 32 bytes of data:
Reply from 192.168.0.200: Destination host unreachable.
Reply from 192.168.0.200: Destination host unreachable.
Reply from 192.168.0.200: Destination host unreachable.
Reply from 192.168.0.200: Destination host unreachable.

Ping statistics for 192.168.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

C:\Users\User1>ping 192.168.0.30

Pinging 192.168.0.30 with 32 bytes of data:
Reply from 192.168.0.30: bytes=32 time=3ms TTL=64
Reply from 192.168.0.30: bytes=32 time=2ms TTL=64
Reply from 192.168.0.30: bytes=32 time=1ms TTL=64
Reply from 192.168.0.30: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.0.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\Users\User1>ping 192.168.0.130

Pinging 192.168.0.130 with 32 bytes of data:
Reply from 192.168.0.130: bytes=32 time<ms TTL=128
Reply from 192.168.0.130: bytes=32 time<ms TTL=128
Reply from 192.168.0.130: bytes=32 time<ms TTL=128
Reply from 192.168.0.130: bytes=32 time<ms TTL=128

Ping statistics for 192.168.0.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

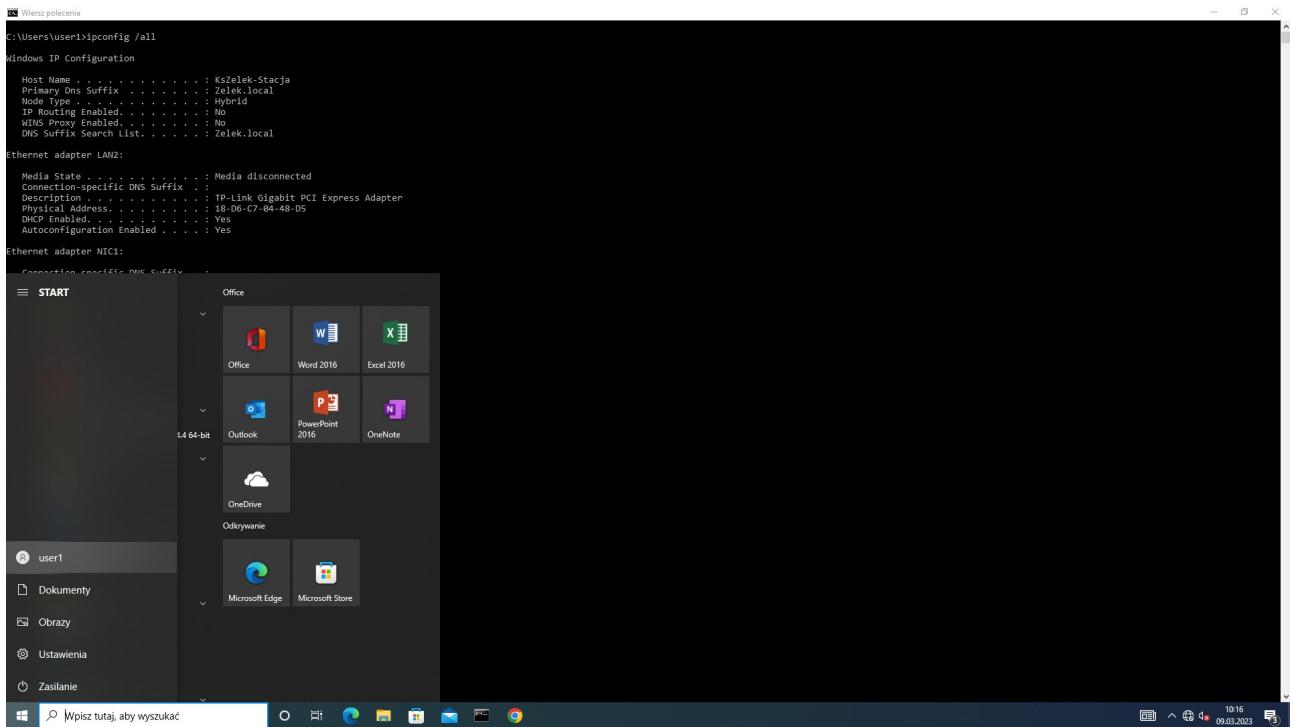
C:\Users\User1>ping 192.168.0.103

Pinging 192.168.0.103 with 32 bytes of data:
Reply from 192.168.0.103: bytes=32 time<ms TTL=64
Reply from 192.168.0.103: bytes=32 time<ms TTL=64
Reply from 192.168.0.103: bytes=32 time<ms TTL=64
Reply from 192.168.0.103: bytes=32 time<ms TTL=64

Ping statistics for 192.168.0.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\User1>
```


12 b.



12 c.

